

Smart India Hackathon 2026

Potential Judge Questions: SkillX Platform

This document contains a curated list of 30 highly specific, non-generic, and challenging questions that SIH judges are likely to ask while directly observing the SkillX web application UI and its underlying architecture.

Category 1: Dashboard UI & Key Metrics

1. "In your dashboard, I see exactly 547 courses split across 36 districts. Is this a hardcoded database, or how did you mathematically arrive at this exact number?"
2. "Your 'Trainees At Risk' metric shows over 1 million trainees. How exactly is this number calculated? Is it a wild assumption or based on real intake capacities?"
3. "Can you explain what the 'Skill Mismatch Index' (72.1%) actually measures? How do you mathematically quantify a 'mismatch'?"
4. "Why is the Course Type Split pie chart divided specifically between ITI and MSSDS? What is the practical difference in how your ingestion engine handles these two catalogues?"
5. "In your Course Alignment Table, why do so many courses have a 'Critical' health status? Isn't it unrealistic that 90% of government courses are critically outdated?"
6. "When I click 'Upgrade Plan' on a course, it opens a Bridge Pack. How long does it take for your system to generate this pack in real-time?"

Category 2: The 4-Engine Architecture & Ideation

7. "What was the specific ideation behind breaking this down into a 4-Engine pipeline rather than just writing a single, monolithic script?"
8. "If the government provides you with a live API tomorrow containing 15,000 active courses, what exactly would you need to change in your codebase to scale this up?"
9. "You mention Engine 2 scrapes job data. How do you ensure you are not hitting rate limits or getting IP banned when scraping hundreds of real job postings?"
10. "Engine 3 extracts skills from unstructured data. How did you handle edge cases where a syllabus mentions 'hands-on practice with Arduino' while a job posting simply asks for 'Microcontroller Programming'? How does the system map these?"
11. "Why did you build this entirely locally without relying on heavy commercial APIs like OpenAI for every single step? What is the long-term cost benefit?"
12. "In Engine 4, how exactly is the alignment score (between 0 and 100) calculated? What specific algorithm or distance metric sits behind this comparison?"

Category 3: The Generated Artifacts (Skill Bridge Packs)

13. "I see the Bridge Pack generated for 'Electrician Trade' recommends 20 hours of additional training. How did the AI determine that it needs exactly 20 hours instead of 50 or 100?"
14. "Who is the primary end-user of this Bridge Pack? Is it the government administrator (for policy change), the institute instructor, or the student themselves?"
15. "Can the Student Portal automatically assign these Bridge Packs to students actively enrolled in a 'Critical Deficit' course?"
16. "Your PDF generation uses PyMuPDF under the hood. Why didn't you just use standard HTML-to-PDF libraries in the frontend, and how does this architectural choice impact your deployment on serverless platforms like Vercel?"
17. "What happens if a newly generated Bridge Pack recommends a highly advanced, emerging skill that the local ITI instructors don't even know how to teach?"

Category 4: State-Wide Intelligence & Industry Demand

18. "Your 'Industry Demand Intelligence' section lists the top demanded skills. Is this data static for the hackathon, or does it update dynamically based on live, incoming jobs?"
19. "How does your system differentiate between the skill demand in an automotive manufacturing hub like Pune versus an agricultural hub like Latur?"
20. "In the District Overview, why does every single district show an average score of less than 75/100? Doesn't the government have at least *some* modernized courses?"
21. "If a specific industry (e.g., EV manufacturing) suddenly surges in Maharashtra next month, how quickly will your dashboard reflect this new localized skill demand?"
22. "You have a feature called 'Skill Dictionary'. Why is a dictionary necessary? Why can't you just compare the raw text of the syllabus directly against the job description?"

Category 5: The Student Portal & Practical Application

23. "I see a Student Portal side to this platform. How does a student interacting with this portal actually bridge their skill gap practically?"
24. "If a student completes the 20-hour Skill Bridge Pack, does your platform offer any certification, or how is their new competency verified to employers?"
25. "Does the Student Portal recommend existing free resources from platforms like NPTEL or Swayam, or is it strictly generating brand new curriculum content via the LLM?"
26. "What prevents a student from simply pasting their syllabus into ChatGPT themselves instead of using your platform? What unique value does SkillIX provide?"

Category 6: System Limitations & Future Scope

27. "Every system has a bottleneck. What is the biggest technical limitation of your current 4-engine pipeline, and how do you plan to fix it if deployed state-wide?"
28. "You rely heavily on LLMs for skill extraction and pack generation. How do you handle 'AI Hallucinations' where the AI might recommend an irrelevant or non-existent tool?"
29. "If you win this hackathon, what is the very first technical feature you would build next before handing this over to the Ministry of Skill Development?"
30. "Ultimately, why should the government adopt SkillIX over just hiring a consulting firm to manually review and update the course syllabi every 3 years?"